CYBERSECURITY LAB PROJECT

Wazuh Windows Endpoint SIEM Monitoring & Detection

PROJECT COMPLETION REPORT

Project	Windows 11 Endpoint Monitoring with Wazuh
SIEM Platform	Wazuh 4.14.8
Endpoint	Windows 11 Pro
Wazuh Agent	Windows11-Lab-01 — Agent ID 004
Endpoint IP	192.168.0.100
Wazuh Server	192.168.0.142
Telemetry	Windows Event Logs + Microsoft Sysmon
Archive Index	wazuh-archives-*
Project Status	COMPLETED

Formal record of the completed Windows-focused Wazuh laboratory phase.

1. Executive Summary

This project phase established and validated a functional Windows endpoint security monitoring pipeline using Wazuh 4.14.8 and Microsoft Sysmon. The Windows 11 endpoint was enrolled as a Wazuh agent, configured to forward Windows event telemetry and Sysmon Operational events, and verified as active in the Wazuh dashboard.

Controlled process activity, including whoami.exe and Notepad, was generated and investigated. Sysmon Event ID 1 process-creation telemetry was observed locally and in Wazuh. Wazuh archive logging and archive indexing were then enabled, allowing raw events to be inspected in Discover.

Completion result: Windows 11 is successfully connected to Wazuh; Sysmon telemetry is collected; Wazuh detections are generated where applicable; and raw archive events are searchable in the dashboard.

2. Objectives Completed

- Deploy and validate a Wazuh 4.14.8 manager.
- Enroll the Windows 11 Pro endpoint as a Wazuh agent.
- Resolve the initial 0.0.0.0 manager-address/enrollment configuration issue.
- Install and verify Microsoft Sysmon.
- Configure the Wazuh agent to ingest Microsoft-Windows-Sysmon/Operational.
- Validate process creation and termination telemetry.
- Investigate a Wazuh Account Discovery detection and its process ancestry.
- Enable Wazuh JSON archives and archive indexing.
- Search and inspect raw events using wazuh-archives-* in Discover.

3. Lab Architecture

Layer	Component	Verified Detail
Endpoint	Windows 11 Pro	192.168.0.100
Telemetry	Microsoft Sysmon	Operational channel; Event ID 1 and Event ID 5 observed
Agent	Wazuh Agent 4.14.8	Windows11-Lab-01; Agent ID 004
Manager	Wazuh 4.14.8	192.168.0.142
Detection	Wazuh Analysis Engine	Rules, alerts and MITRE mapping
Storage	Wazuh Archives	wazuh-archives-*
Investigation	Wazuh Discover	Raw fields and full_log inspected

4. Implementation & Configuration

4.1 Windows Agent Enrollment

The Windows endpoint was enrolled against 192.168.0.142. The unique agent name Windows11-Lab-01 was used after the initial name was rejected as non-unique. The invalid manager address 0.0.0.0 was corrected to 192.168.0.142, and an enrollment configuration was added for port 1515, agent name and default group. After restart, the agent connected successfully.

4.2 Agent Validation

The Wazuh dashboard showed Agent 004 as ACTIVE, version 4.14.8, IP 192.168.0.100, running Windows 11 Pro. Application, Security and System event monitoring was confirmed.

4.3 Sysmon Installation

Sysmon was downloaded from Microsoft Sysinternals, extracted and installed. The Sysmon64 service was verified as Running. Local EventChannel queries confirmed Sysmon was generating process telemetry.

4.4 Sysmon Collection by Wazuh

The Wazuh agent configuration was extended with a localfile entry for Microsoft-Windows-Sysmon/Operational using the eventchannel log format. The Wazuh service was restarted and its log confirmed analysis of the Sysmon channel.

Field	Observed Value
Image	C:\Windows\System32\whoami.exe
Process ID	36000
User	CYB3RGWD\PC
Integrity Level	High
Parent Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Parent Process ID	44144
Channel	Microsoft-Windows-Sysmon/Operational
Event ID	1 — Process Create

5. Detection & Investigation

5.1 Account Discovery Detection

A Wazuh detection named Discovery activity executed was investigated. The command was net user and Wazuh mapped it to MITRE ATT&CK; T1087 — Account Discovery under the Discovery tactic.

Process ancestry showed approximately: wazuh-agent.exe → net.exe → net1.exe → net user. Because the parent was wazuh-agent.exe running as SYSTEM and the working directory was the Wazuh agent directory, the evidence was consistent with Wazuh-generated/legitimate lab telemetry. This exercise demonstrated why SOC analysts must investigate process ancestry and context rather than treating a detection label alone as proof of malicious activity.

5.2 whoami.exe Test

5.3 Notepad Test

Notepad was launched with Start-Process notepad.exe as a benign process-creation test. Sysmon recorded Event ID 1 with executable path, user, integrity level, parent PowerShell process, process ID and hash information. The corresponding raw telemetry was confirmed at the Wazuh server archive layer.

6. Wazuh Archive & Discover Validation

Archive logging was enabled on the Wazuh manager using the global settings for logall and logall_json. The configuration test completed successfully and the Wazuh manager restarted successfully.

Archive indexing was then enabled so wazuh-archives-* could be selected in Discover. The archive index displayed more than 54,000 events during the observed seven-day period, confirming that raw

telemetry was being indexed.

The resulting data proved the complete path: Windows 11 → Sysmon → Wazuh Agent → Wazuh Manager → wazuh-archives-* → Discover.

7. Results & Evidence

Validation	Result
Wazuh manager	ACTIVE / running
Windows agent	ACTIVE — Agent 004
Endpoint	192.168.0.100
Manager	192.168.0.142
Sysmon64	RUNNING
Sysmon channel	Collected by Wazuh Agent
Event ID 1	Observed and archived
Event ID 5	Observed and archived
Account Discovery	Observed; MITRE T1087
whoami.exe	Observed in wazuh-archives-*
Notepad	Observed locally and in server archives
Archive index	wazuh-archives-* working in Discover

8. SOC / Blue-Team Skills Demonstrated

- Endpoint onboarding and Wazuh agent troubleshooting.
- Windows EventChannel collection and validation.
- Sysmon installation and endpoint telemetry generation.
- Process creation and termination investigation.
- Parent-child process ancestry analysis.
- MITRE ATT&CK; technique interpretation.
- Distinguishing alerts from raw archived events.
- JSON event-field investigation in Wazuh Discover.
- End-to-end SIEM telemetry validation.

9. Completion Statement

The Windows-focused Wazuh laboratory phase is complete. The environment successfully demonstrates endpoint telemetry collection, transport, storage, detection and investigation. The Windows endpoint is ready to remain as the first monitored host while the lab proceeds to the next virtual machine.

PROJECT STATUS: COMPLETED — WINDOWS WAZUH/SYSMON MONITORING PHASE

10. Next Phase

Proceed to the Kali Linux virtual machine and onboard it to the existing Wazuh environment, followed by additional controlled laboratory hosts and security telemetry.

End of Report

KALI LINUX + WAZUH FIM

Project Analysis Appendix — Integrated with the completed Windows 11 Wazuh/Sysmon phase

Project platform: Wazuh 4.14.8

Wazuh Manager: 192.168.0.142

Kali endpoint: cyb3rgwd — Agent ID 005 — 192.168.0.193

Monitoring capability validated: File Integrity Monitoring (FIM/Syscheck) with real-time detection

1. Executive Summary

This phase extended the previously completed Windows 11 Wazuh/Sysmon laboratory by onboarding the Kali Linux endpoint to the same Wazuh Manager and validating real-time File Integrity Monitoring. The Kali Wazuh Agent service was confirmed active, and controlled file operations were generated against `/wazuh-test/fim-test-02.txt`.

Wazuh successfully detected the file creation, modification and deletion events and presented them in the Wazuh dashboard. The modification event also recorded integrity changes across size, timestamp, MD5, SHA1 and SHA256 values. This provides a second monitored operating system and demonstrates the end-to-end SIEM workflow across Windows and Linux endpoints.

2. Combined Lab Architecture

Layer	Windows 11 Endpoint	Kali Linux Endpoint
Host / Agent	Windows11-Lab-01 / ID 004	cyb3rgwd / ID 005
IP address	192.168.0.100	192.168.0.193
Telemetry	Windows Event Logs + Sysmon	Syscheck/FIM + Linux logs
Wazuh Manager	192.168.0.142	192.168.0.142
Validated controls	Process creation, Sysmon, archives	File create / modify / delete

3. Kali FIM Validation

The following controlled operations were performed on the Kali endpoint.

Test	Controlled action	Wazuh result
File creation	Created /wazuh-test/fim-test-02.txt	Detected as added
File modification	Wrote new test content to the same file	Detected as modified; integrity checksum chang
File deletion	Removed /wazuh-test/fim-test-02.txt	Detected as deleted; rule description: File delete

4. Modification Event Analysis

The dashboard returned one matching event for the query `agent.name:cyb3rgwd AND rule.groups:syscheck AND syscheck.event:modified`.

The event showed decoder **syscheck_integrity_changed** and rule description **Integrity checksum changed.** The event identified the file as `/wazuh-test/fim-test-02.txt` and reported changes to size, modification time, MD5, SHA1 and SHA256. The event was processed in realtime mode.

5. Deletion Event Analysis

A second dashboard query filtered for `syscheck.event:deleted` and returned one hit. The expanded document identified decoder **syscheck_deleted**, rule description **File deleted**, and the same monitored path. The Wazuh rule level shown in the event was 5.

6. SOC Interpretation

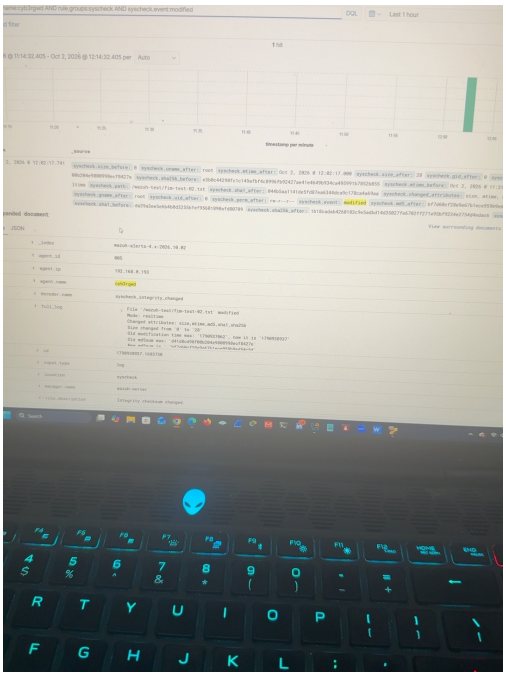
These tests demonstrate the core FIM analyst workflow: establish a monitored asset, perform a controlled change, observe the generated telemetry, validate the event fields, and correlate the change with the affected path and integrity metadata. The result is evidence that the Kali endpoint is not merely enrolled but is actively producing security telemetry that reaches the central Wazuh analysis and dashboard layers.

The events were deliberately generated as benign laboratory activity. Their presence therefore demonstrates detection capability rather than evidence of compromise.

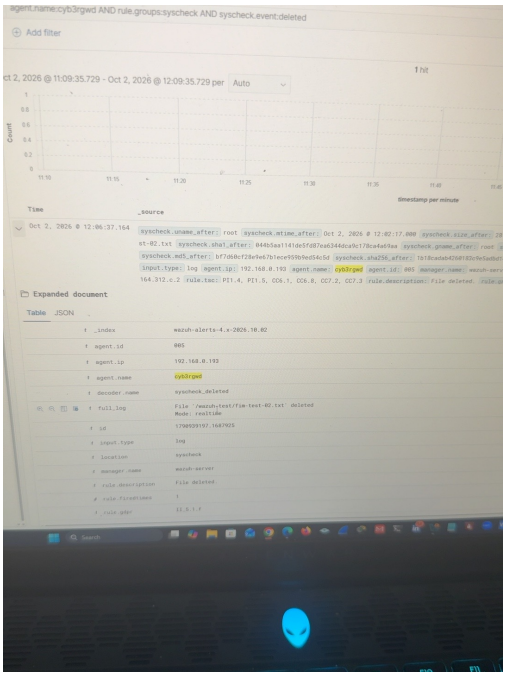
7. Evidence & Operational Notes

The Kali terminal showed the Wazuh agent service in an active/running state. The file modification command returned the expected test string, and the dashboard subsequently displayed the corresponding Syscheck event. The deletion test likewise produced a dashboard event for the same path.

Evidence snapshot — FIM modification event



Dashboard result: syscheck.event = modified; integrity checksum changed.



Dashboard result: syscheck.event = deleted; file deletion detected.

8. Minor Troubleshooting Observations

During the lab, a Wazuh DQL query was briefly typed into the Kali shell, which correctly returned `command not found`. This was only a command-context mistake: DQL belongs in the Wazuh dashboard search bar, not the Linux terminal. It did not affect the Wazuh agent or FIM configuration.

9. Combined Windows + Kali Project Assessment

The original Windows-focused phase established centralized monitoring of a Windows 11 Pro endpoint using Wazuh Agent 4.14.8 and Microsoft Sysmon. That phase validated process-creation telemetry, MITRE ATT&CK; mapping, archive logging, and raw-event investigation in Discover.

The Kali phase now adds Linux endpoint telemetry to the same Wazuh Manager and validates real-time FIM through controlled file creation, modification and deletion. Together, the two phases demonstrate multi-platform endpoint visibility and centralized security-event analysis.

10. Consolidated Validation Matrix

Capability	Windows 11	Kali Linux
Wazuh enrollment	Validated — Agent 004	Validated — Agent 005
Central manager	192.168.0.142	192.168.0.142
Endpoint telemetry	Validated	Validated
Process telemetry	Sysmon Event ID 1/5	Not part of this FIM test
FIM / Syscheck	Available through Wazuh	Validated: add/modify/delete
Dashboard investigation	Validated	Validated
Raw/archive investigation	Validated	Central event path validated

11. Current Project Status

Windows 11 Wazuh/Sysmon phase: completed and documented in the original report.

Kali Linux Wazuh/FIM phase: completed and documented in this appendix.

Next planned lab phase: onboard Metasploitable3 and continue controlled security telemetry and detection exercises.

PROJECT STATUS: WINDOWS + KALI WAZUH MONITORING VALIDATED

This combined document preserves the original Windows report and appends the Kali Linux FIM analysis as a single project record.